Infosys System Engineer: Mock Test — Paper 7

Total questions: 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

Section	Questions	Time
1. Reasoning Ability	Q1–15	25 min
2. Technical Ability (Quant)	Q16–25	35 min
3. Verbal Ability	Q26–45	20 min
4. Pseudo Code	Q46–50	10 min
5. Numeric Puzzle Solving	Q51–54	10 min
6. English Grammar	Q55–59	10 min
7. English Writing	Q60	10 min

Topic coverage map

Section	Topics covered in this paper
Reasoning	Syllogism, data sufficiency, coding-decoding, seating arrangement, data interpretation, blood relations, direction sense, number series, mirror image, ranking
Quant	Permutations, probability, profit and loss, time and work, boats and streams, percentages, compound interest, remainders, ratio and ages, LCM
Verbal	Reading comprehension, critical reasoning, sentence correction, para-jumbles, sentence completion, synonyms and antonyms
Pseudo Code	Increment/decrement operators, recursion, graph traversal, modular arithmetic loops, dynamic programming
Puzzles	Matchstick, circular wheel, number combination (cryptarithm), matrix logic
Grammar	Subject-verb agreement, tenses, prepositions, error spotting, articles

SECTION 1: Reasoning Ability (15 Q, 25 min)

Q1. Statements: Some servers are clients. All clients are nodes. No node is offline. Some offline things are routers.

Conclusions: I. Some servers are not offline. II. Some routers are not nodes. III. No client is a router. IV. Some nodes are servers.

(a) I and II only (b) I, II and IV only (c) I, II, III and IV (d) II, III and IV only

Q2. *Data Sufficiency.* What is the value of x + y?

I. $x^2 + 2xy + y^2 = 49$

II. x and y are both positive.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

Q3. *Data Sufficiency.* Is the average of the set S greater than 10?

I. The median of S is 12.

II. S contains exactly 5 positive integers.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

Q4. In a code, each letter is replaced by its opposite letter in the alphabet (A↔Z, B↔Y, C↔X, and so on), and the resulting word is then written in reverse order. Using this rule, CODE is coded as VWLX. What is the code for SYSTEM?

(a) NVGHBH (b) HBHGVN (c) NVGHBG (d) MVGHBH

Q5. In a certain code, ROSE is written as 6821, CHAIR is written as 73456, and PREACH is written as 961473. How is SEARCH written in that code?

(a) 214673 (b) 214763 (c) 241673 (d) 214637

Questions 6 and 7 are based on the following information.

Eight people A, B, C, D, E, F, G and H sit around a circular table facing the centre. A sits opposite E. B sits immediately to the right of A. C sits second to the left of B. D sits immediately to the right of E. G sits opposite C. F sits second to the right of A.

Q6. Who sits opposite B?

(a) D (b) H (c) F (d) G

Q7. Who sits third to the left of F?

(a) C (b) A (c) H (d) B

Questions 8 and 9 are based on the following table.

Year	Revenue (₹ crore)	Expenditure (₹ crore)
2021	420	350
2022	510	400
2023	600	480
2024	660	550
2025	780	600

(Profit = Revenue − Expenditure.)

Q8. Over the five years together, total profit is what percentage of total expenditure (to the nearest whole number)?

(a) 25% (b) 24% (c) 26% (d) 28%

Q9. Between which two consecutive years was the percentage growth in revenue the lowest?

(a) 2021–2022 (b) 2022–2023 (c) 2023–2024 (d) 2024–2025

Q10. A is the mother of B. B is the sister of C. C is the father of D. E is the only brother of A. How is E related to D?

(a) Paternal grand-uncle (b) Maternal uncle (c) Paternal uncle (d) Grandfather

Q11. A ship sails 20 km on a bearing of 030°, and then 20 km on a bearing of 330°. What is its final position relative to the starting point?

(a) 20√3 km (about 34.6 km) due North (b) 20 km due North (c) 40 km due North (d) 20√3 km due East

Q12. Find the next term: 1, 3, 12, 60, 360, ?

(a) 2520 (b) 2160 (c) 1800 (d) 2400

Q13. Find the next term: 3, 8, 6, 14, 12, 26, ?

(a) 24 (b) 22 (c) 50 (d) 52

Q14. The image of a wall clock in a plane mirror shows the time as 8:25. What is the actual time?

(a) 3:35 (b) 4:35 (c) 3:25 (d) 4:25

Q15. In a row of 40 students, A is 15th from the left and B is 18th from the right. How many students are between A and B?

(a) 7 (b) 6 (c) 8 (d) 9

SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

Q16. In how many ways can the letters of the word SUCCESS be arranged so that no two S's are adjacent?

(a) 120 (b) 60 (c) 240 (d) 420

Q17. A fair coin is tossed 6 times. What is the probability that no two heads appear consecutively?

(a) 21/64 (b) 13/32 (c) 1/3 (d) 5/16

Q18. A shopkeeper marks his goods 40% above cost, gives a 15% discount on the marked price, and also uses a weight of 950 g in place of 1 kg. What is his actual profit percentage (approximately)?

(a) 25.26% (b) 19% (c) 24% (d) 26.3%

Q19. A and B together can do a job in 12 days, B and C together in 15 days, and C and A together in 20 days. In how many days can A alone do the job?

(a) 30 days (b) 24 days (c) 20 days (d) 40 days

Q20. A boat takes 6 hours to go 24 km upstream and 28 km downstream, and 6½ hours to go 30 km upstream and 21 km downstream. What is the speed of the boat in still water?

(a) 10 km/h (b) 8 km/h (c) 12 km/h (d) 9 km/h

Q21. In an examination, 70% of the candidates passed in Mathematics, 60% passed in Science, and 15% failed in both subjects. What percentage of the candidates passed in both subjects?

(a) 45% (b) 40% (c) 50% (d) 55%

Q22. The difference between the compound interest (annual compounding) and the simple interest on a sum for 3 years at 20% per annum is ₹640. What is the sum?

(a) ₹5,000 (b) ₹4,000 (c) ₹6,000 (d) ₹5,500

Q23. What is the remainder when 2^{2026} is divided by 7?

(a) 2 (b) 1 (c) 4 (d) 3

Q24. The present ages of A and B are in the ratio 5 : 3. After 4 years, the ratio of their ages will be 3 : 2. What is the sum of their present ages?

(a) 32 years (b) 36 years (c) 40 years (d) 28 years

Q25. Three bells ring at intervals of 12, 15 and 20 minutes. They all ring together at 9:00 AM. How many times in all (including 9:00 AM) will they ring together between 9:00 AM and 3:00 PM, both times included?
(a) 7 (b) 6 (c) 8 (d) 5

SECTION 3: Verbal Ability (20 Q, 20 min)

Read the passage and answer Q26–30.

> *In distributed systems, a network call that times out leaves the caller in genuine ignorance: the request may have been lost before reaching the server, executed successfully with the reply lost on the way back, or still be in flight. Retrying is the natural response, but retrying a non-idempotent operation — charging a card, say — risks performing it twice. The standard remedy is to make the operation idempotent by attaching a unique request identifier, so that the server can recognise a duplicate and return the original result instead of re-executing. This shifts the burden rather than removing it: the server must now durably remember identifiers, and must decide how long to remember them, since storing them forever is costly while discarding them too early reopens the window for duplicates. Engineers who treat idempotency as a mere implementation detail therefore tend to discover it as an incident, whereas those who treat it as part of an operation's contract design it in from the start.*

Q26. The passage mainly argues that:
(a) Network timeouts can be eliminated with better hardware
(b) Safe retries need idempotency, which has its own costs and is best designed in from the start
(c) Card payments should never be retried
(d) Servers should discard request identifiers immediately

Q27. "Genuine ignorance" in the first sentence refers to the fact that:
(a) the caller does not know the server's address
(b) the server refuses to reply
(c) the caller cannot tell whether the request was executed
(d) the network is always unreliable

Q28. The author says the identifier approach "shifts the burden rather than removing it" because:
(a) the caller must now generate more traffic
(b) the server must store identifiers and decide how long to keep them
(c) the identifiers make the operation non-idempotent
(d) duplicates become impossible to detect

Q29. According to the passage, what is the risk of discarding identifiers too early?
(a) A duplicate request may be executed a second time
(b) The server will run out of storage
(c) The caller will stop retrying
(d) The original result will be corrupted

Q30. Which inference is best supported by the last sentence?
(a) Idempotency is only needed in payment systems
(b) Incidents are caused only by implementation details
(c) Teams that include idempotency in an operation's contract are less likely to meet it first as an outage
(d) Contracts should not describe retry behaviour

Critical Reasoning (Q31–33).

Q31. After a website adopted a new CDN, its page-load time fell by 30%, and user retention rose over the same period. The team concludes that the CDN caused the rise in retention. Which of the following is an assumption on which this conclusion depends?

(a) Retention is determined only by page-load time

(b) No other factor that affects retention changed significantly over the same period

(c) All CDNs improve retention

(d) A 30% fall in load time is always considered large

Q32. Since a company made code reviews mandatory, production defects have fallen by 40%. The company concludes that code reviews reduce defects. Which of the following, if true, most weakens this conclusion?

(a) Each review takes about two hours a week

(b) In the same period the company halved its release frequency and cut new-feature work sharply

(c) Developers say they like the review process

(d) The defect-tracking system was not changed

Q33. A start-up claims its new caching layer will cut its database costs. Which of the following, if true, most strengthens the claim?

(a) In a pilot, 85% of requests were served from the cache, and every cache hit avoids a billable database query

(b) The caching layer was designed by a well-known engineer

(c) The start-up's database costs rose last year

(d) The caching layer is built on open-source software

Sentence Correction (Q34–36).

Q34. The data, which was collected over three years, shows that **fewer people has adopted** the system than expected.

(a) No improvement (b) fewer people have adopted (c) less people have adopted (d) fewer people is adopting

Q35. The throughput of the new cluster is much higher **than the old one**.

(a) No improvement (b) than that of the old one (c) than the old one's (d) from the old one

Q36. I wish the system **was** more reliable than it is now.

(a) No improvement (b) were (c) is (d) would be

Para-jumbles (Q37–39).

Q37.

P: This is why eventually consistent replicas can briefly serve stale reads.

Q: In a replicated database, a write is first applied to one node and then propagated to the others.

R: This window may be small, but it is never zero.

S: During propagation, different nodes hold different versions of the same record.

(a) QRSP (b) QSRP (c) SQRP (d) QSPR

Q38.

P: Hash tables achieve average constant-time lookup.

Q: Collisions, however, are inevitable.

R: Chaining and open addressing are the two standard ways to resolve them.

S: If many keys land in the same bucket, performance degrades towards linear time.

(a) PQRS (b) PQSR (c) QPSR (d) PSQR

Q39.

P: That decision, in hindsight, saved the company millions.

Q: In 2019 the board chose to rewrite its payment platform rather than patch it.

R: Patching would have cost less that year.

S: The rewrite also delayed several product launches.

(a) QRSP (b) QSRP (c) RQSP (d) QSPR

Sentence Completion (Q40–41).

Q40. Although the prototype looked impressive in the demo, its performance under real load was _____.

(a) unimpeachable (b) underwhelming (c) exemplary (d) gratifying

Q41. The engineer's explanation was so _____ that even the non-technical audience grasped the idea at once.

(a) convoluted (b) lucid (c) esoteric (d) verbose

Vocabulary, Synonyms and Antonyms (Q42–45).

Q42. Synonym of **ABSTRUSE**:

(a) obvious (b) difficult to understand (c) lengthy (d) outdated

Q43. Synonym of **SALIENT**:

(a) minor (b) hidden (c) most noticeable or important (d) rejected

Q44. Antonym of **ARDUOUS**:

(a) strenuous (b) demanding (c) effortless (d) lengthy

Q45. Antonym of **OBSEQUIOUS**:

(a) servile (b) fawning (c) assertive (d) flattering

SECTION 4: Pseudo Code (5 Q, 10 min)

Q46. What is printed? (Operands are evaluated strictly from left to right; `x++` and `x--` return the old value and then change x, while `++x` and `--x` change x first and then return the new value.)

```
x = 4
y = x++ * 2 + ++x
z = --x + x-- * 2
print(x + y + z)
```

(a) 33 (b) 31 (c) 35 (d) 29

Q47. What is printed? (/ is integer division)

```
int g(int n) {
    if (n == 0) return 0;
    if (n % 2 == 1) return g(n - 1) + n;
    return g(n / 2) + n / 2;
}
print(g(10))
```

(a) 14 (b) 17 (c) 15 (d) 12

Q48. The graph below is stored as an adjacency list, and this iterative traversal marks a node as visited only when it is popped. In what order are the nodes visited?

```
adj[0] = [1, 2]
adj[1] = [3, 4]
adj[2] = [4, 5]
adj[3] = []
adj[4] = [6]
adj[5] = [6]
adj[6] = []

stack = [0]
while (stack is not empty)
    v = pop(stack)
    if (v is visited) continue
    mark v visited and print v
    for each w in adj[v] (in the order listed)
        push(stack, w)
    end for
end while
```

(a) 0 2 5 6 4 1 3 (b) 0 1 3 4 6 2 5 (c) 0 1 2 3 4 5 6 (d) 0 2 4 6 5 1 3

Q49. What is printed?

```
arr = [3, 4, 10, 11, 14, 7, 0]
count = 0
for i = 0 to 6
    for j = i + 1 to 6
        if ((arr[i] + arr[j]) % 7 == 0)
            count = count + 1
        end if
    end for
end for
print(count)
```

(a) 7 (b) 6 (c) 8 (d) 5

Q50. This is the 0/1 knapsack dynamic programme. What is printed?

```
wt  = [2, 3, 4, 5]
val = [3, 4, 5, 6]
W = 5
dp[0..W] = 0
for i = 0 to 3
    for w = W down to wt[i]
        dp[w] = max(dp[w], dp[w - wt[i]] + val[i])
    end for
end for
print(dp[W])
```

(a) 7 (b) 6 (c) 8 (d) 9

SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

Q51. *Matchstick.* A 3 × 3 grid of unit squares is built from 24 matchsticks and contains 14 squares in total (nine 1×1, four 2×2 and one 3×3). What is the minimum number of sticks that must be removed so that exactly 2 complete squares remain?

(a) 4 (b) 3 (c) 5 (d) 6

Q52. *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 3, R = 4, B = 5, L = 2, Centre = 9

Wheel 2: T = 6, R = 2, B = 4, L = 5, Centre = 17

Wheel 3: T = 7, R = 3, B = 2, L = 6, Centre = ?

(a) 14 (b) 18 (c) 12 (d) 20

Q53. *Number combination.* In the addition SEND + MORE = MONEY, each letter stands for a different digit (0–9), and S and M are not zero. What is the value of D + E + Y?

(a) 14 (b) 12 (c) 16 (d) 13

Q54. *Matrix logic.* Find the missing number.

3	4	5
5	6	19
7	8	?

(a) 41 (b) 39 (c) 43 (d) 47

SECTION 6: English Grammar (5 Q, 10 min)

Q55. A series of failures _____ the board to reconsider the entire strategy.

(a) have led (b) has led (c) are leading (d) were led

Q56. By the time the client called, the team _____ the fix for several hours.

(a) tests (b) had been testing (c) has tested (d) was tested

Q57. The new policy is incompatible _____ our existing contracts, so we must opt _____ it, or we will be held liable _____ any breach.

(a) with, out of, for (b) to, out from, of (c) with, out off, to (d) to, out of, for

Q58. *Error spotting.* "The number of vulnerabilities discovered this year (A) / are significantly higher (B) / than that of last year, (C) / according to the report. (D)"

(a) A (b) B (c) C (d) D

Q59. She is _____ MBA from _____ university in _____ UK and works as _____ HR manager.

(a) an, a, the, an (b) a, an, the, a (c) an, an, -, an (d) a, a, the, an

SECTION 7: English Writing (1 Q, 10 min)

Q60. *Write a short technical essay/paragraph of 150–180 words.*

Explain how the "round robin" and "least connections" load-balancing algorithms differ. Then recommend which one is better for a service whose requests vary greatly in how long they take to complete, and justify your recommendation.

Evaluation rubric (10 marks): Technical accuracy of the two algorithms (3) | Soundness and justification of the recommendation (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)

ANSWER KEY WITH EXPLANATIONS

Section 1: Reasoning

1. **(b)** Some servers are clients and clients are nodes, which are never offline, so those servers are not offline (I). Some offline things are routers, and nodes are never offline, so those routers are not nodes (II). Some servers are nodes, which converts to "some nodes are servers" (IV). III does not follow, because other routers (not the offline ones) might be clients.

2. **(c)** I gives $(x + y)^2 = 49$, so $x + y = 7$ or -7. II alone gives no value. Together, $x + y$ must be positive, so $x + y =$ **7**.

3. **(d)** A set such as {1, 1, 12, 12, 12} has median 12 and 5 positive integers but an average of 7.6, while {12, 12, 12, 13, 100} has an average above 10. Even together, the statements are not sufficient.

4. **(a)** S→H, Y→B, S→H, T→G, E→V, M→N gives HBHGVN, which reversed is **NVGHBH** (check: CODE→XLWV→VWLX ✓).

5. **(a)** From the three words: R=6, O=8, S=2, E=1, C=7, H=3, A=4, I=5, P=9. SEARCH = S(2) E(1) A(4) R(6) C(7) H(3) = **214673**.

6. **(a)** Fix A at one seat and number the seats clockwise from A: A0, C1, H2, D3, E4, G5, F6, B7. (E is opposite A at 4; B is to A's right at 7; C is second to B's left at 1; D is to E's right at 3; G is opposite C at 5; F is second to A's right at 6; H takes the last seat.) The seat opposite B (7) is seat 3, which is **D**.

7. **(a)** F is at seat 6. Moving to the left (clockwise): first is B (7), second is A (0), third is **C** (1).

8. **(a)** Profits: 70, 110, 120, 110, 180, totalling 590. Total expenditure = 350+400+480+550+600 = 2380. 590/2380 ≈ 24.8%, which is **25%**.

9. **(c)** Growth: 2021–22 = 21.4%, 2022–23 = 17.6%, 2023–24 = 10%, 2024–25 = 18.2%. The lowest is **2023–2024**.

10. **(a)** A is the mother of B, and B is C's sister, so A is also C's mother. C is D's father, so A is D's paternal grandmother. E is A's only brother, so E is D's **paternal grand-uncle**.

11. **(a)** Leg 1: East = $20 \sin 30° = 10$, North = $20 \cos 30° = 10\sqrt{3}$. Leg 2 (bearing 330°): East = -10, North = $+10\sqrt{3}$. The East parts cancel; the North parts add to $20\sqrt{3} \approx 34.6$ km, so the ship is **$20\sqrt{3}$ km due North**.

12. **(a)** Each term is the previous term multiplied by 3, 4, 5, 6 and then 7: 360 × 7 = **2520**.

13. **(a)** Two interleaved series. Odd positions: 3, 6, 12, then ×2 gives 24. Even positions: 8, 14, 26 (each is double the previous minus 2). The 7th term, in an odd position, is **24**.

14. **(a)** Actual time = 11:60 − 8:25 = **3:35**.

15. **(a)** B is 18th from the right, so B is 40−18+1 = 23rd from the left. Students between A (15th) and B (23rd) = 23−15−1 = **7**.

Section 2: Quant

16. **(a)** Arrange the non-S letters U, C, C, E first: 4!/2! = 12 ways. This creates 5 gaps, and the 3 S's go into 3 different gaps: C(5,3) = 10. Total = 12×10 = **120**.

17. **(a)** The number of binary strings of length 6 with no two consecutive heads is the Fibonacci number 21, out of 2^6 = 64. Probability = **21/64** (confirmed by listing all 64 outcomes).

18. **(a)** Take cost per kg = 100. Marked price = 140; after a 15% discount the selling price for a claimed 1 kg is 119. The customer actually gets only 950 g, which costs the shopkeeper 95. Profit = 119 − 95 = 24 on a cost of 95, so 24/95 ≈ **25.26%**.

19. **(a)** (A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 12/60 = 1/5, so A+B+C = 1/10. A = 1/10 − (B+C) = 1/10 − 1/15 = 1/30, so A alone needs **30 days**.

20. **(a)** Let upstream speed = u and downstream speed = d. 24/u + 28/d = 6 and 30/u + 21/d = 6.5. Solving gives u = 6 and d = 14 (check: 4 + 2 = 6 ✓, 5 + 1.5 = 6.5 ✓). Speed in still water = (6+14)/2 = **10 km/h**.

21. **(a)** Passed in at least one subject = 100 − 15 = 85%. Both = 70 + 60 − 85 = **45%**.

22. **(a)** For 3 years at 20%, CI − SI = $P(1.2^3 - 1 - 0.6) = 0.128P = 640$, so P = **₹5,000**.

23. **(a)** $2^3 = 8 \equiv 1 \pmod 7$, so powers of 2 repeat every 3. 2026 = 3×675 + 1, so $2^{2026} \equiv 2^1 = \mathbf{2}$.

24. **(a)** Let the ages be 5x and 3x. (5x+4)/(3x+4) = 3/2 gives 10x + 8 = 9x + 12, so x = 4. The ages are 20 and 12, and the sum is **32**.

25. **(a)** LCM(12, 15, 20) = 60 minutes. In 6 hours (360 minutes) the bells ring together at 0, 60, 120, 180, 240, 300 and 360 minutes: **7 times**.

Section 3: Verbal

26. **(b)** 27. **(c)** 28. **(b)** 29. **(a)** 30. **(c)**

31. **(b)** The conclusion that the CDN caused the retention rise only holds if nothing else that affects retention changed over the same period.

32. **(b)** Halving releases and cutting feature work gives an alternative explanation for fewer defects, so it weakens the claim that reviews were the cause.

33. **(a)** A high cache-hit rate, with each hit avoiding a billable query, directly links the cache to lower database costs.

34. **(b)** "Fewer" is used with countable nouns (people), and the plural subject "people" takes "have".

35. **(b)** Compare like with like: the throughput of the new cluster with **that of** the old one.

36. **(b)** After "wish", a present unreal condition takes the subjunctive "were".

37. **(b)** Q→S→R→P. 38. **(b)** P→Q→S→R. 39. **(a)** Q→R→S→P.

40. **(b)** 41. **(b)**

42. **(b)** 43. **(c)** 44. **(c)** 45. **(c)**

Section 4: Pseudo Code

46. **(a) 33.** Line 2: x++ returns 4 (x becomes 5); ×2 = 8; ++x makes x = 6 and returns 6; y = 8 + 6 = 14, x = 6. Line 3: --x makes x = 5 and returns 5; x-- returns 5 (x becomes 4); 5×2 = 10; z = 5 + 10 = 15, x = 4. Result = 4 + 14 + 15 = **33**.

47. **(a) 14.** g(1)=g(0)+1=1; g(2)=g(1)+1=2; g(4)=g(2)+2=4; g(5)=g(4)+5=9; g(10)=g(5)+5 = **14** (confirmed by code).

48. **(a)** Visit 0 and push 1, 2. Pop 2, visit, push 4, 5. Pop 5, visit, push 6. Pop 6, visit. Pop 4, visit (its neighbour 6 is pushed but already visited). Pop 6 (skipped), pop 1, visit, push 3, 4. Pop 4 (skipped), pop 3, visit. Order: **0 2 5 6 4 1 3**.

49. **(a) 7.** Taking each value modulo 7 gives 3, 4, 3, 4, 0, 0, 0. A pair sums to a multiple of 7 when it is (3,4) or (0,0). There are 2×2 = 4 pairs of a 3 with a 4, and C(3,2) = 3 pairs of zeros, so the count is 4 + 3 = **7** (confirmed by code).

50. **(a) 7.** The best choice for capacity 5 is items of weight 2 and 3, worth 3 + 4 = **7**. The loop runs w downwards, so each item is used at most once.

Section 5: Numeric Puzzles

51. **(a) 4.** An exhaustive computer search over all 24 sticks shows that no set of 3 sticks leaves exactly 2 squares, while a set of 4 does. Minimum = **4**.

52. **(a) 14.** Rule: Centre = (T × B) + (R × L) − (T + R + B + L). Wheel 1: 15 + 8 − 14 = 9 ✓. Wheel 2: 24 + 10 − 17 = 17 ✓. Wheel 3: 14 + 18 − 18 = **14**.

53. **(a) 14.** The only solution is S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2 (9567 + 1085 = 10652). D + E + Y = 7 + 5 + 2 = **14**.

54. **(a) 41.** Rule: third number = (first × second) − (first + second). Row 1: 12 − 7 = 5 ✓. Row 2: 30 − 11 = 19 ✓. Row 3: 56 − 15 = **41**.

Section 6: Grammar

55. **(b)** "A series" is a singular subject, so the verb is "has led".

56. **(b)** An activity that was in progress for some time before a past event takes the past perfect continuous: "had been testing".

57. **(a)** incompatible **with**, opt **out of**, liable **for**.

58. **(b)** The subject is "the number" (singular), so the verb should be "is", not "are".

59. **(a)** an MBA (the sound /em/), a university (the sound /ju/), the UK (a specific country name), an HR manager (the sound /eɪtʃ/).

Section 7: Writing

Use the rubric above. A strong answer explains that round robin sends requests to servers in a fixed rotation regardless of their current load, while least connections sends each new request to the server with the fewest active connections. For requests of widely varying duration, least connections is the better choice, because round robin can pile long requests onto one server while others sit idle.

Scoring guide

Award +1 per MCQ (Q1–59) and up to 10 for Q60. This is the hardest paper in the series. The seating arrangement, probability, matchstick, cryptarithm and all pseudo-code answers were verified by running the logic in code.